

CISTERN-7 wet-well cross-section -- role start/stop elevations by deadband setting

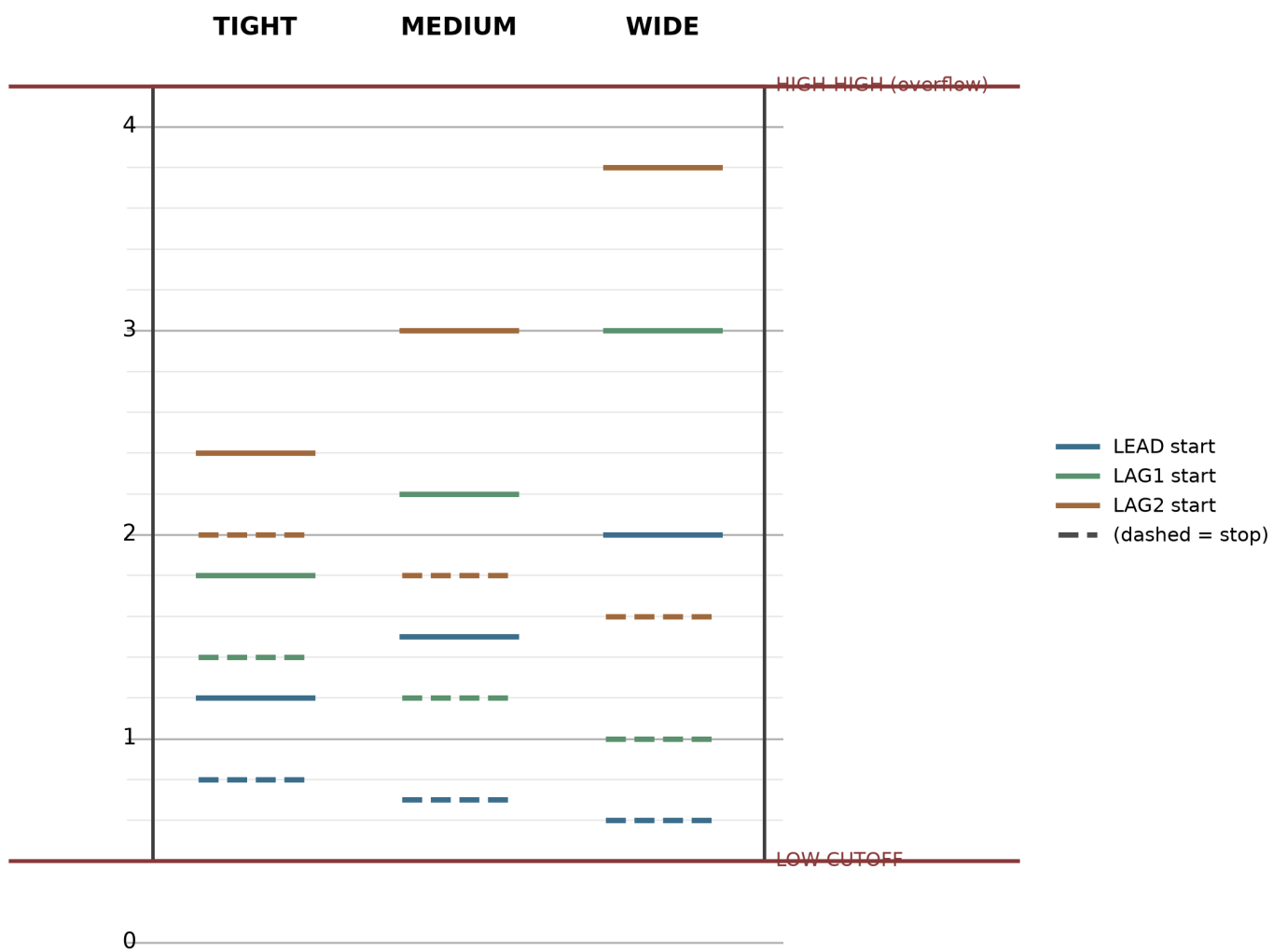


Figure 1 -- Wet-well cross-section. Role start/stop elevations by deadband setting.

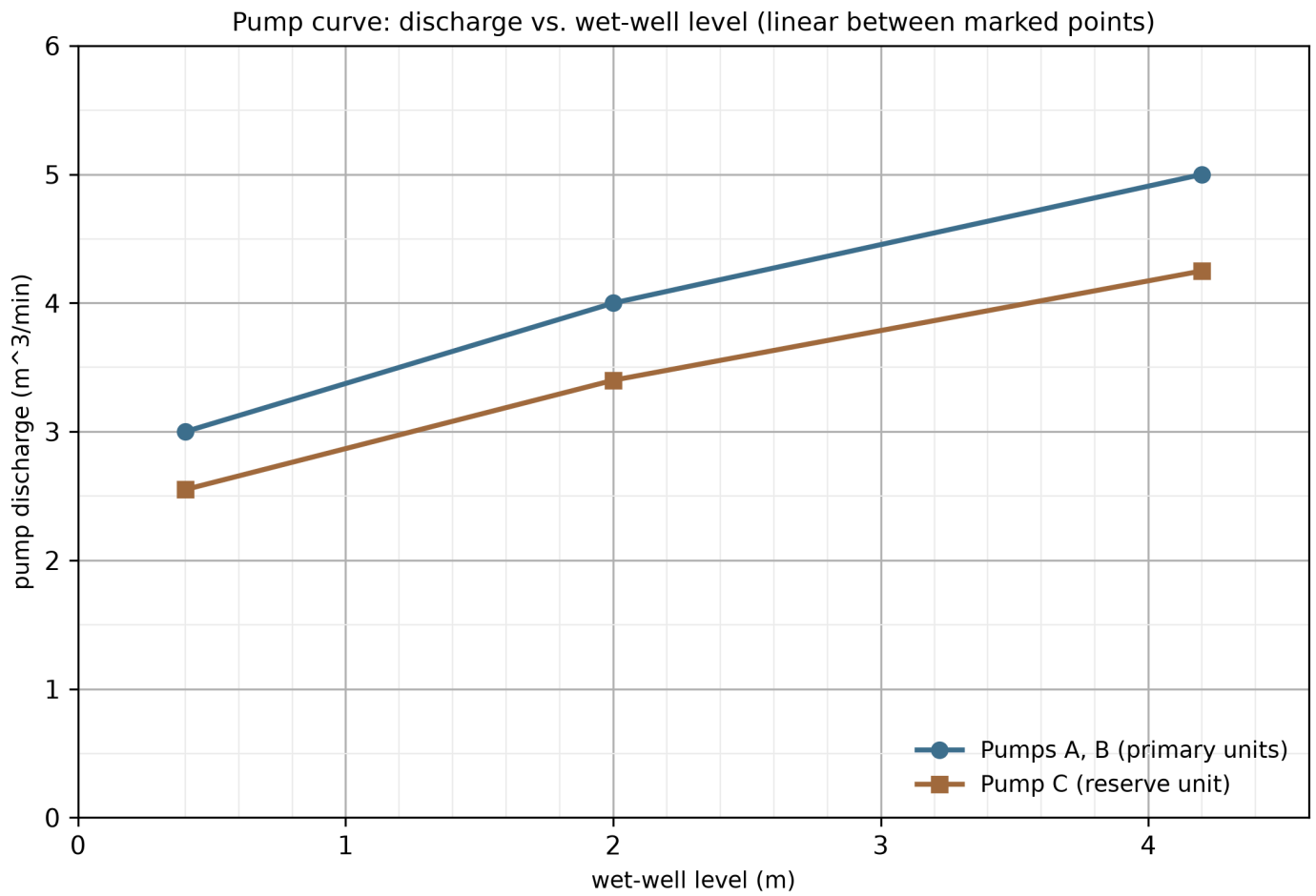


Figure 2 -- Pump curve: discharge vs. level. Compare A/B's curve to C's to read C's scale.

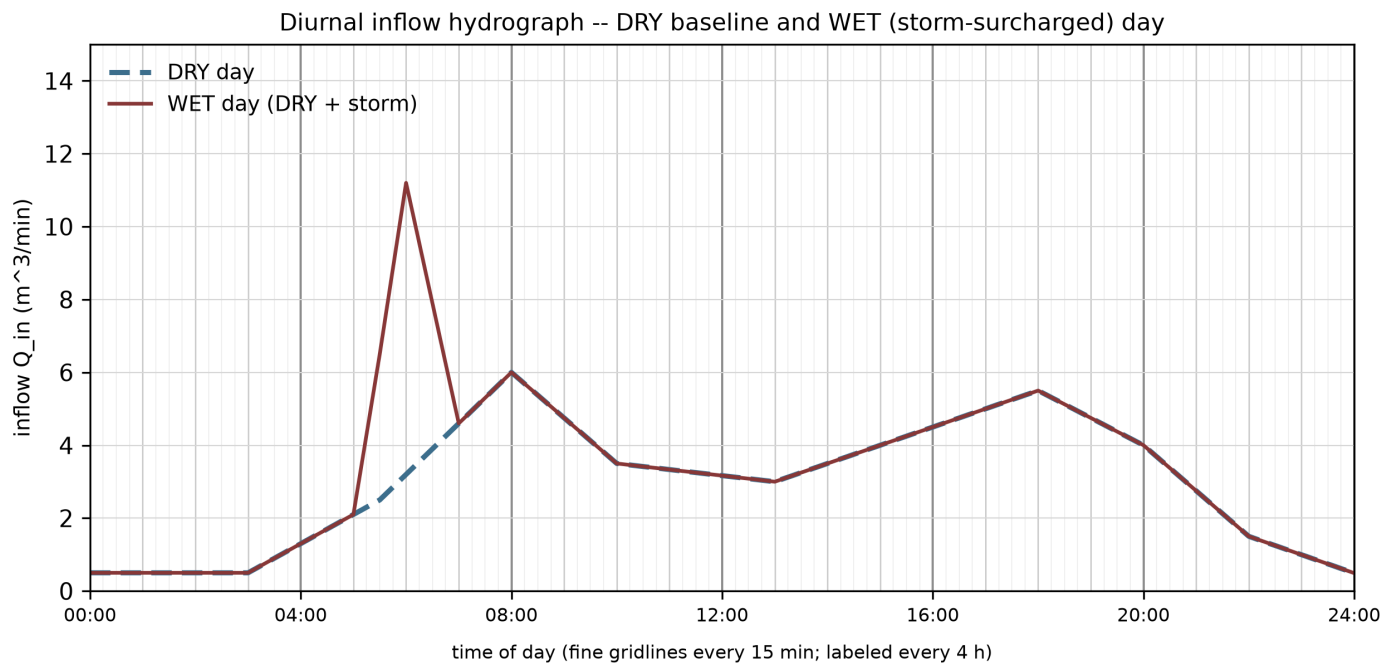


Figure 3 -- Diurnal inflow hydrograph, DRY and WET days.

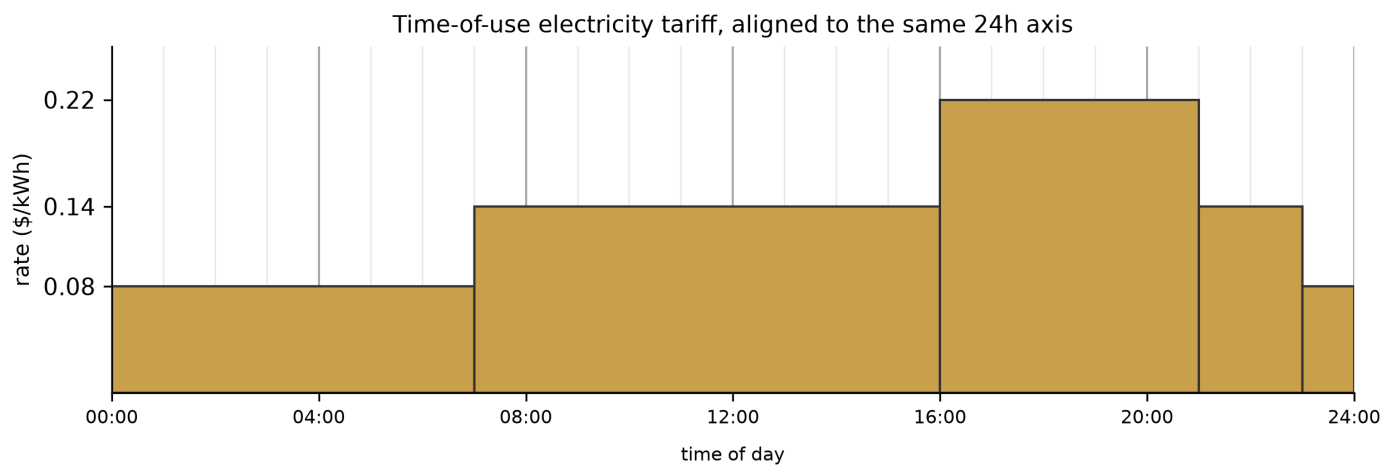


Figure 4 -- Time-of-use tariff, aligned to the same 24h axis as Figure 3.

1. SIMULATION HORIZON

One simulated day is 1440 one-minute ticks, $t=0$ (00:00) through $t=1439$ (23:59), run to completion -- never a shortened or sampled run. The wet well is a vertical tank of constant plan area 60.0 m^2 . $\text{level}(t)$ is a single continuous state, updated each tick from that tick's own inflow and pump discharge (see Figures 3, 2): $\text{level}(t+1) = \text{level}(t) + (Q_{\text{in}}(t) - Q_{\text{out}}(t)) / 60.0$, clamped to $[0.4, 4.2] \text{ m}$ (Figure 1). Whenever the unclamped update would exceed 4.2 m , the excess volume (times 60.0 m^2) is a spill added to overflow_volume for that tick, and level is clamped to exactly 4.2 m .

2. INFLOW (Figure 3)

$Q_{\text{in}}(t)$ for a DRY day is the plotted DRY curve, linearly interpolated between its marked anchor points. A WET day adds the plotted storm surcharge on top of the same DRY curve; the surcharge is a triangle, zero outside its marked window, linearly ramping to its labeled peak and back to zero. Q_{in} is never a discrete arrival list -- it is this continuous function, sampled once per tick.

3. PUMP CURVE AND STAGING (Figures 1, 2)

Three pumps A, B, C. Each running pump's discharge is interpolated from Figure 2 against the current $\text{level}(t)$; pump C's curve is scaled relative to pumps A/B's (read the exact ratio from Figure 2 at any level). A non-running pump discharges zero. Each pump fills one of three roles each tick -- LEAD, LAG1, LAG2 -- assigned fresh each tick: LEAD holds whichever pump is currently offered that role (rotation policy, item 5); the remaining in-roster pumps fill LAG1 then LAG2 in ascending pump-ID order. A role's pump starts when $\text{level}(t)$ reaches that role's own start elevation (Figure 1) and it is currently off; it stops when $\text{level}(t)$ falls to that role's own stop elevation and it is currently on, EXCEPT that a stop is deferred while the pump is blocked by its minimum-run timer (item 4). Only the lowest-ID duty_pump_count pumps (a sweep knob: 1, 2, or 3) are ever in service; pump C is the first excluded as the roster shrinks. A pump outside the roster never runs.

4. PER-PUMP TIMERS

Minimum run time (a sweep knob: SHORT=8 or LONG=15 ticks): once a pump starts, it cannot stop for that many ticks even if level has already reached its stop elevation; it keeps running (and keeps discharging) until the timer clears, then the stop condition is re-checked every tick. Minimum off time (fixed, 4 ticks, not swept): once a pump stops, it cannot restart for 4 ticks. A pump blocked from stopping by its minimum-run timer still counts toward duty_pump_count for staging additional pumps.

5. LEAD ROTATION POLICY (a sweep knob)

The lead role is offered to a pump only at the instant no pump currently fills LEAD and $\text{level}(t)$ reaches the would-be lead's own start elevation. The rotation index advances by exactly one position IMMEDIATELY AFTER a genuine LEAD-START transition completes -- never on a LAG start, never on any stop event, and never merely because a tick elapsed. Three policies: STRICT_ALTERNATE cycles the in-roster pumps in fixed ID order; RUNTIME_BALANCED offers it to the in-roster pump with the lowest cumulative running minutes so far (ties to lowest ID); FIXED_LEAD always offers it to the lowest-ID in-roster pump (no rotation).

6. ENERGY AND TARIFF (Figure 4)

Each running pump draws a constant 15 kW, independent of level. Each tick's energy cost is (running pump count) * 15 kW * (1/60 h) * the tariff rate whose band (Figure 4) contains that tick's own start minute. `total_energy_cost` is the sum over all 1440 ticks.

7. THE SEARCH -- 54-configuration sweep

`day_type` is weather, not a control setting -- it is never a free dimension a selection optimizes over. The legal configuration space is `duty_pump_count` x `rotation_policy` x `deadband` x `min_run_time` = {1,2,3} x {STRICT_ALTERNATE, RUNTIME_BALANCED, FIXED_LEAD} x {TIGHT, MEDIUM, WIDE} x {SHORT, LONG} = 54 configurations; deadband elevations are fixed per setting (Figure 1) and not independently swept. Every configuration is evaluated as a PAIR of full 1440-tick runs, one per `day_type` -- the sweep table has 54 x 2 = 108 rows, but a selection is always over the 54 configurations, scored by both of a configuration's own two rows together, never by picking whichever day happens to be cheaper. For configuration k : $E(k)$ = combined DRY+WET energy cost; $P(k)$ = worse-day peak level; $\text{imbalance}(k)$ = worse-day (highest single pump's run-minutes / that day's total fleet run-minutes); feasible iff `overflow_volume` is zero on both days. Report three lexicographic selections:

- energy-optimal: among feasible configurations, minimize $E(k)$
- reliability-optimal: minimize $P(k)$ over ALL configurations, tie-break lower $E(k)$
- wear-balance-constrained: among feasible configurations with $\text{imbalance}(k) \leq 0.80$, minimize $E(k)$

These three selections are not required to agree.

8. CERTIFICATION

For the baseline configuration (`duty_pump_count=3`, `rotation_policy=STRICT_ALTERNATE`, `deadband=TIGHT`, `min_run_time=LONG`, `day_type=WET`) only: serialize the complete per-tick trace as one line per tick "`t|level|q_in|q_out|running_pumps|lead`" (level/q_in/q_out to exactly 6 decimal places; `running_pumps` as comma-separated 0-indexed pump numbers in ascending order, `A=0/B=1/C=2`, empty string if none running; `lead` as the 0-indexed pump number or the literal text `None`), preceded by one header line "`CISTERN7|(3, 'STRICT_ALTERNATE', 'TIGHT', 'LONG', 'WET')`" (Python tuple repr of the five config values in that order), lines joined by `\n`, UTF-8 encoded, hashed with SHA-256; report the first 16 hex characters as the trace-integrity certificate.

Deliver: simulator source, an independently-coded verifier, the full 108-row sweep table, a decision file with all three selections, the baseline trace plus its SHA-256 certificate and your adversarial-mutation results, and an engineering memo. Base every reported number on your own executed implementation.